

Cloze Encounters: The Impact of Pirated Data Access on LLM Performance

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Quick Overview

- Research Question: effect of data access on LLM performance
- Data Access: Inference on whether a book appears in *Books3*
- LLM Performance: measured by *Name Cloze*, directly captures memorization, arguably related to general intelligence or model utility (Petroni et al., 2019)

Original text (fiction-style):

The door swung open and Sarah stepped into the quiet library. She glanced around until she spotted Daniel sitting by the window, completely absorbed in his book. Without saying a word, she walked over and tapped him lightly on the shoulder.

Name-cloze version:

*The door swung open and Sarah stepped into the quiet library. She glanced around until she spotted **[MASK]** sitting by the window, completely absorbed in his book. Without saying a word, she walked over and tapped him lightly on the shoulder.*

- Within the *Name-Cloze* setting:
 - What baseline accuracy would you expect an LLM to achieve on a directly trained book?
 - What baseline accuracy would you expect an LLM to achieve on an indirectly exposed (e.g. Wikipedia summaries) book?
 - How much would you expect accuracy to increase if the model has only indirect exposure (e.g. Wikipedia summaries) but not the full text?
 - How much would you expect accuracy to increase if the model has no prior exposure?

- Start with a universe of books from ISBNdb (794k titles collected using keyword searches across publication years **1800–2023**).
- Match these books to **Goodreads** to obtain metadata (519k matched).
- Identify which books belong to the *Books3* pirated dataset using fuzzy matching of titles and authors (196k potential treatment books).
- Retrieve full text from **Libgen**; only books with accessible text can be used for cloze tasks (16k texts found).
- Generate 40–60 token Name-Cloze passages and 100 passages for most books using NER (15k books successfully processed).
- Query multiple LLMs on all cloze passages (14k books with complete responses across models).
- Excluding **mismatches**, get 12,916 books.

Data: Discussion on Data Construction

- Period: 1800-2023 v.s. 1990-2022

Why choose 1800-2023?

- Goodreads and Lebgen Selection?

Books with metadata from Goodreads and Lebgen are endogenously more popular

- Mismatch

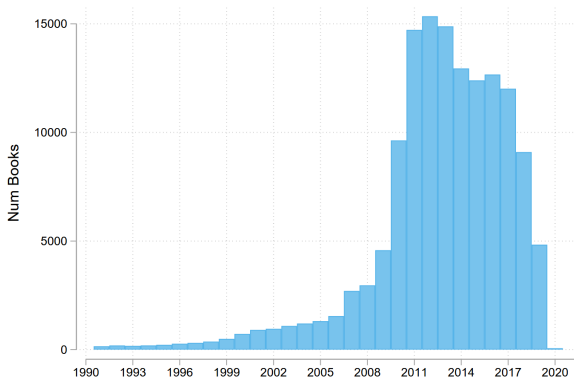
What kind of mismatch could happen?

Data: *Book3*

- **Books3:** 196k pirated books (Presser 2022), likely used in LLM pretraining.
- **Year distribution:** Large spike in 2011–2014 (based on 196k).

Figure 2: Books3 Data Distribution by Publication Year

(a) Books3 Publication Years (195,997 books)



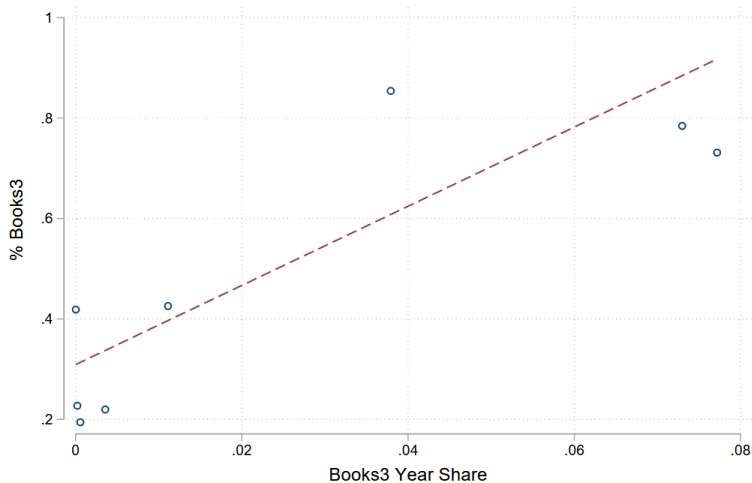
Data: Summary Statistics

Table 1: Summary Statistics

Statistic	Mean	St. Dev.	Median	Min	Max	N
Book Characteristics						
Books3	0.49	0.50	0	0	1	12,916
Total ratings	15,515.35	154,672.60	87	0	6,147,566	12,916
Total reviews	656.67	4,956.55	9	0	178,631	12,916
Average rating	3.55	1.16	3.87	0.00	5.00	12,916
Publication year	1,989.16	48.97	2,004	1,000	2,023	12,886
Fiction	0.29	0.45	0	0	1	9,562
0-10 ratings	0.24	0.43	0	0	1	12,916
10-100 ratings	0.28	0.45	0	0	1	12,916
100-1000 ratings	0.25	0.43	0	0	1	12,916
1000+ ratings	0.23	0.42	0	0	1	12,916
LLM Results						
GPT 3.5 Turbo	0.12	0.11	0.09	0.00	1.00	12,916
GPT 4o	0.11	0.11	0.08	0.00	1.00	12,916
Claude	0.14	0.12	0.12	0.00	1.00	12,916
Llama 8B	0.04	0.05	0.02	0.00	0.67	12,916
Llama 70B	0.15	0.13	0.12	0.00	1.00	12,916
Gemini	0.16	0.13	0.13	0.00	1.00	12,916

Note: This table presents summary statistics for the main data samples used in this study. For each book, we collected data from Goodreads on various characteristics like total number of ratings, total number of reviews, etc. Roughly half of the books in the sample are in Books3. Only 9,562 of the 12,916 in the sample have genre listed and of the 9,562 close to 70% are nonfiction.

- Problem: Books3 inclusion is endogenous to corpora popularity.
- Solution: Use instrumental variables to isolate quasi-exogenous variation in Books3 inclusion.
- IV: Share of Books3 titles published in year t ($Share_t$).
- Key Fact: Books3 has a highly uneven publication-year profile, providing variation in LLM access to data.
- Claim: $Share_t$ strongly predicts Books3 inclusion;
Conditional on popularity controls, Publication year should not directly affect LLM Name-Cloze accuracy.



Design: Discussion on IV

- **Exclusion Restriction:** PublishYear is still likely correlated to popularity;
- **Control Group:** How can we be sure that books outside *Books3* are not included in other training corpora (e.g., CommonCrawl, wikis, summaries, reviews)? This threatens the validity of the control group.
- **Publication-Year Effects:** Older and newer books differ in linguistic style, readability, and name-predictability. If language drift or genre evolution affects Name-Cloze difficulty, publication year may have a direct effect on accuracy, violating exclusion.
- **Goodreads Selection:** Goodreads coverage is itself a function of digital adoption over time. Books published later may have more online presence independent of Books3, creating a year–popularity confound.

Results

Figure 3: Model-free Evidence: Publication Year vs Accuracy across LLMs

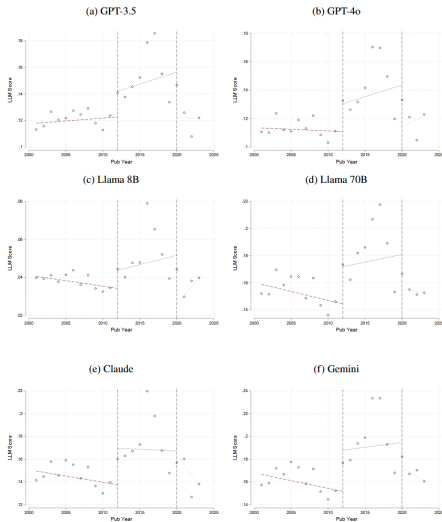


Table 2: Baseline IV Regressions

	(1) GPT3.5 Turbo	(2) GPT 4o	(3) Llama 8b	(4) Llama 70b	(5) Claude	(6) Gemini
Books3	0.0284*** (0.00437)	0.0231*** (0.00422)	0.000796 (0.00183)	0.0133*** (0.00502)	0.0104** (0.00492)	0.0110** (0.00521)
N	12916	12916	12916	12916	12916	12916
F-Stat	3890.6	3890.6	3890.6	3890.6	3890.6	3890.6

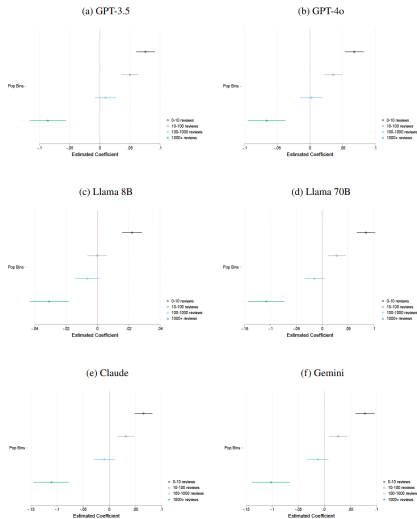
Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: This table presents estimates from the main IV regression as described in the text at the book level. The main outcome variable is the average accuracy score on the cloze task at the book level for each of the six models studied in the paper. The F-Stat presented is from the first stage regression where Books3 membership for book from publication year t is instrumented with the share of Books3 books from year t of the total number of books in that dataset. Robust standard errors are presented.

Results

Figure 4: Books3 Effect By Book Popularity



Note: This figure summarizes estimates from 24 IV regressions estimating the effect of Books3 inclusion for four different popularity bins for each model in separate samples.

- Why are effects negative for highly popular books?
- Is *Name Cloze* a good metric? other metrics on *Memorization* and *Generalization*?
- If we value content only by its marginal contribution to LLM performance, is that a fair basis for compensating content creators?